

patrn.ink — Feature Reference Guide

1. Authentication

Endpoint	Method	Description
/auth/google/login	GET	Redirects to Google OAuth consent screen
/auth/github/login	GET	Redirects to GitHub OAuth consent screen
/api/me	GET	Returns current user profile

Auth Flow:

1. User clicks login → redirected to provider (Google/GitHub)
2. Provider redirects back to API callback
3. API creates/updates user, generates JWT
4. API redirects to frontend: {FRONTEND_URL}/auth/callback?token={JWT}
5. Frontend stores JWT in localStorage, uses Authorization: Bearer {token} header

User Model:

```
{
  id:      string    // UUID
  email:   string
  name:    string
  picture: string     // Avatar URL
  provider: string    // "google" or "github"
  created_at: string  // ISO 8601
}
```

2. Link Management

Create Link

POST /api/shorten

```
{
  "long_url":      "https://example.com/very/long/url", // Required
  "custom_code":   "my-link",                          // Optional, 3-20 alphanumeric/hyphen
  "expires_in":    168,                                // Optional, hours until expiration
  "scheduled_at":  "2026-04-01T...",                   // Optional, ISO 8601 activation time
  "tags":          ["portfolio"],                       // Optional, string array
  "password":      "secret123",                         // Optional, will be bcrypt-hashed
  "title":         "My Portfolio",                      // Optional
  "description":   "Personal site",                     // Optional
  "age_verification": 0                                // Optional: 0=none, 1=13+, 2=18+, 3=21+
}
```

Response:

```
{
  "short_url": "https://patrn.ink/abc123",
  "short_code": "abc123",
  "long_url": "https://example.com/very/long/url",
  "qr_code_url": "https://patrn.ink/abc123/qr",
  "expires_at": "2026-04-08T...",
  "scheduled_at": null,
  "tags": ["portfolio"]
}
```

List Links (Paginated)

GET /api/links

Query Param	Type	Description
search	string	Search in short_code, long_url, title
tags	string[]	Filter by tags (repeat param)
page	int	Page number (default: 1)
limit	int	Items per page (default: 10)
sort_by	string	created_at OR clicks
sort_order	string	asc OR desc
archived	bool	Show only archived links

Response:

```
{
  "links": [...],
  "total": 42,
  "page": 1,
  "limit": 10,
  "total_pages": 5
}
```

Get Link Detail

GET /api/links/:code

Update Link

PUT /api/links/:code

Accepts same fields as create (except `custom_code`), plus:

- `is_archived` : bool — soft archive/unarchive

Delete Link

DELETE /api/links/:code

Link Model

```
{
  short_code:      string
  long_url:        string
  user_id:         string
  custom_alias:    bool      // true if user chose the code
  clicks:          int       // total click count
  created_at:      string    // ISO 8601
  expires_at:      string?   // optional
  scheduled_at:    string?   // optional
  is_active:       bool
  is_archived:     bool
  tags:            string[]?
  password:        string?   // present = protected (hashed)
  title:           string?
  description:      string?
  age_verification: int      // 0=none, 1=13+, 2=18+, 3=21+
}
```

3. Analytics

Get Analytics Summary

GET /api/analytics/:code

Query Param	Type	Description
start_date	string	ISO date (YYYY-MM-DD)
end_date	string	ISO date (YYYY-MM-DD)

Response:

```
{
  "total_clicks": 1234,
  "unique_clicks": 890,
  "top_referrers": { "google.com": 200, "twitter.com": 150 },
  "clicks_by_date": { "2026-03-20": 45, "2026-03-21": 62 },
  "clicks_by_hour": { "0": 5, "1": 3, ... "23": 12 },
  "device_types": { "Desktop": 600, "Mobile": 500, "Tablet": 134 },
  "browser_types": { "Chrome": 700, "Safari": 300, "Firefox": 234 },
  "countries": { "US": 400, "UK": 200, "IN": 150 },
  "timeline": [{ "date": "2026-03-20", "clicks": 45 }, ...]
}
```

Export Analytics

GET /api/analytics/:code/export

Query Param	Type	Values
format	string	csv OR json
start_date	string	ISO date
end_date	string	ISO date

Returns downloadable file.

4. Bulk Operations

Bulk Delete / Archive

POST /api/bulk/delete

```
{
  "codes": ["abc123", "xyz789"],
  "archive": true // true = archive, false = permanent delete
}
```

Response:

```
{
  "deleted": ["abc123", "xyz789"],
  "failed": { "bad1": "not found" } // optional
}
```

Bulk Import

POST /api/bulk/import

```
{
  "links": [
    { "long_url": "https://example.com", "custom_code": "ex", "tags": ["test"], "title": "Example" },
    { "long_url": "https://other.com" }
  ]
}
```

Response:

```
{
  "created": [ { "short_url": "...", "short_code": "..." } ],
  "failed": [ { "index": 1, "url": "...", "reason": "invalid URL" } ]
}
```

Export All Links

GET /api/export/links?format=csv|json

5. API Tokens

Create Token

POST /api/tokens

```
{
  "name":      "My CLI Token",
  "scopes":    ["links:read", "links:write", "analytics:read", "bulk:read", "bulk:write"],
  "expires_in": 30    // Optional, days
}
```

Response:

```
{
  "token": "ptk_abc123...full_token_shown_once",
  "api_token": {
    "id":      "uuid",
    "name":    "My CLI Token",
    "token_prefix": "ptk_abc1",
    "scopes":  ["links:read", "links:write"],
    "rate_limit": 60,
    "created_at": "2026-03-24T...",
    "is_active": true
  }
}
```

Important: The full `token` value is only returned at creation. Store it immediately.

List Tokens

GET /api/tokens → { tokens: APIToken[] }

Revoke Token

DELETE /api/tokens/:id

Update Rate Limit

PUT /api/tokens/:id/rate-limit

```
{ "rate_limit": 120 }
```

Available Scopes

Scope	Access
links:read	Read links
links:write	Create/update/delete links

analytics:read	View analytics
bulk:read	Export links
bulk:write	Bulk import/delete

6. Public / Redirect Endpoints

Redirect

GET `/:code`

If unprotected → HTTP 302 redirect to `long_url`

If protected → returns JSON:

```
{ "password_required": true }  
// or  
{ "age_required": "18+", "age_level": 2 }
```

Verify Password

POST `/:code/verify`

```
{ "password": "user-entered-password" }
```

→ { "redirect_url": "https://..." }

Verify Age

POST `/:code/verify-age`

```
{ "confirmed": true, "age_level": 2 }
```

→ { "redirect_url": "https://..." }

QR Code

GET `/:code/qr` → PNG image of QR code

Link Preview (by code)

GET `/:code/preview`

```
{  
  "title":      "Page Title",  
  "description": "Meta description",  
  "image":      "https://og-image.jpg",  
  "favicon":    "https://favicon.ico",  
  "url":        "https://example.com",  
}
```

```
"domain":      "example.com"
}
```

URL Preview (by raw URL)

```
GET /api/preview?url=https://example.com
```

Same response as above.

7. UI Pages to Implement

Route	Purpose
/	Landing page with OAuth login
/auth/callback	Receives ?token= from API, stores JWT, redirects to dashboard
/dashboard	Overview: stats cards, quick-create input, recent links
/dashboard/links	Links table with search, sort, filter, pagination, bulk select
/dashboard/links/[code]	Link detail: all metadata, QR code, inline analytics chart, edit/delete
/dashboard/analytics	Link selector + date range → timeline chart, device/browser pie charts, referrer bar chart, CSV/JSON export
/dashboard/tokens	Create token (name, scopes, expiry), list with prefix/scopes/status, revoke
/dashboard/settings	User profile display, theme toggle, sign out
/[code]	Public gate: password input form or age confirmation prompt

8. Tech Stack

Layer	Technology
Backend	Go, Gin framework
Database	AWS DynamoDB
Cache	Redis
Frontend	Next.js 16, TypeScript, Tailwind CSS v4
Auth	OAuth 2.0 (Google, GitHub) + JWT

Generated from [patrn.ink](#) API source code analysis — March 2026